

TrES-2b

R-2182

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_111013 · created 2026-08-13T22:53:12+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455847.65459 +/- 0.00011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.019
Transit depth (Rp/Rs)^2	1.16 +/- 0.4 [%]
Orbital Inclination (inc)	84.25 +/- 1.16
Airmass coefficient 1 (a1)	1.203 +/- 0.01
Airmass coefficient 2 (a2)	-0.1401 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	0.83 %
Best Comparison Star	None
Optimal Aperture	3.67
Optimal Annulus	5.97
Transit Duration (day)	0.073 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.108 ± 0.019 vs 0.1254 (deviation 0.85σ)
Duration fit vs published	0.073 d vs 0.0746 d (deviation 0.07σ)
Mid-time O–C	-0.2 min
Depth SNR	5.3
Residual scatter	0.83 %

Light curve

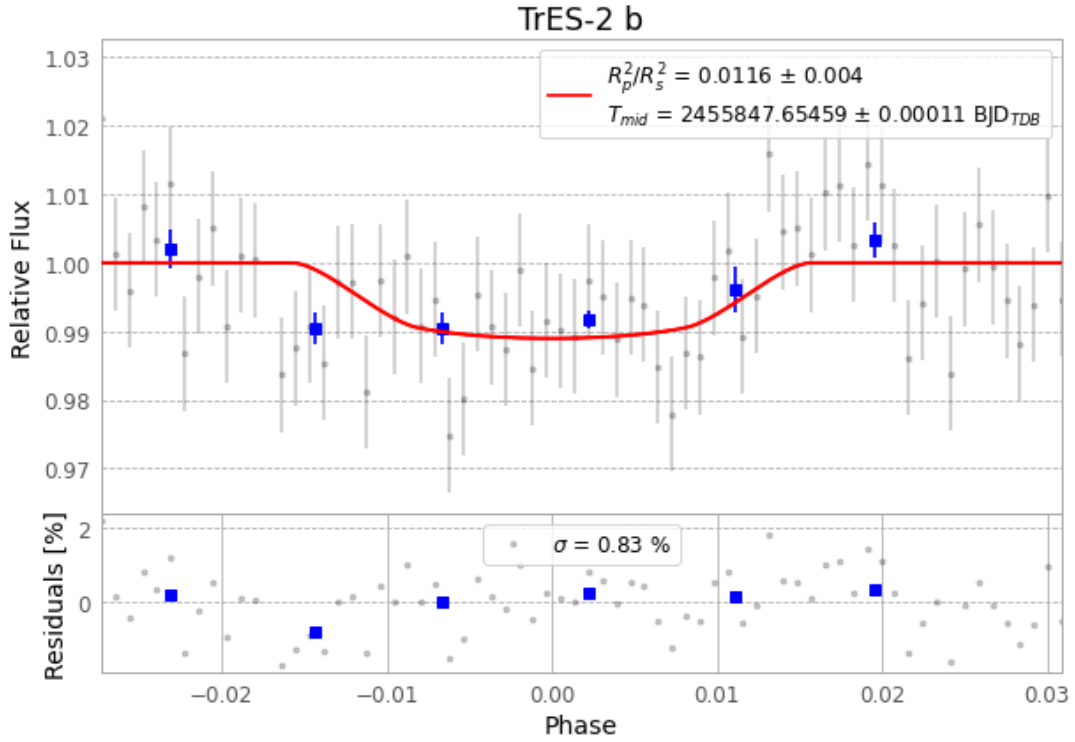


Figure 1 — FinalLightCurve_TrES-2 b_2011-10-12.png (SHA-256 7c1ee723508a31b0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8cfc61c3349f3e7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), TRES-2111013020312.FITS → TRES-2111013053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-111013.log (1f4d0418b411...), FinalLightCurve_TrES-2 b_2011-10-12.png (7c1ee723508a...), FinalLightCurve_TrES-2 b_2011-10-12.csv (7b47e5262e48...)

Compute resources

Wall time 102.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 22:53Z.